

Water Management at IIT Gandhinagar

IIT Gandhinagar runs an integrated water system that saved about 266.5 million litres (ML) in FY 2025–26, through rainwater harvesting, decentralised sewage treatment, reuse and smart monitoring.

Overview

IIT Gandhinagar is India's first 5-star GRIHA LD campus. It is a zero-discharge campus: there is no outgoing municipal sewage line, and all sewage is treated on site and reused for landscape irrigation. Together, rainwater harvesting, eco-friendly sewage treatment (DEWATS) and smart monitoring have reduced annual water demand by 30%.

The strategy works in four layers:

- 1. **Dual-source supply:** Narmada canal water plus harvested rainwater, treated centrally.
- 2. **Digital and mechanical guardrails:** metering, automated controls and leak response.
- 3. **Zero-discharge wastewater treatment:** a chemical-free, zero-energy DEWATS.
- 4. **Reuse:** treated wastewater irrigates the campus landscape.

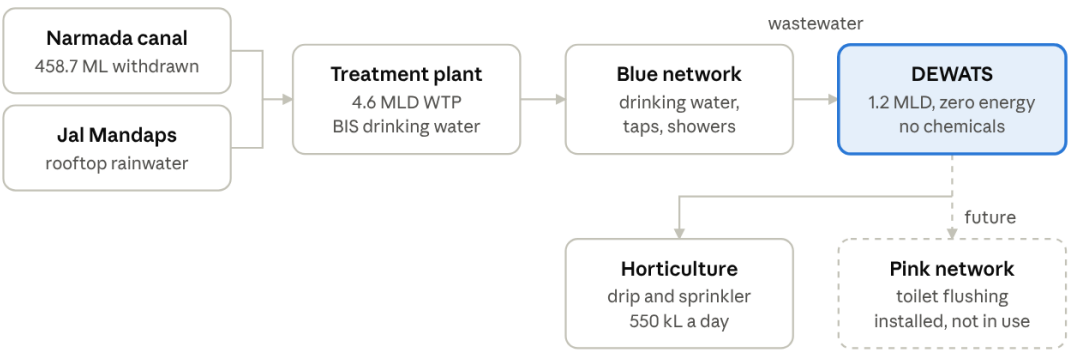
Water Sources

Narmada canal water is the primary supply, and harvested rainwater supplements it.

Source	Role	Quantity, FY 2025–26
Narmada canal	Primary, high-volume surface water for central treatment	458.7 ML withdrawn
Rooftop rainwater (Jal Mandaps)	Secondary; also dilutes turbid canal water in the monsoon	30 ML collected

Campus Water Cycle

Treated wastewater replaces fresh water for landscape irrigation



Fresh water is treated once for drinking and washing; the resulting wastewater is treated again at DEWATS and used for landscape irrigation instead of leaving the campus. A separate recycled-water (pink) network for toilet flushing is installed but not yet in use.

Water Treatment Plant

The 4.6 MLD Water Treatment Plant (WTP) supplies potable water continuously and currently runs at an average of 1.2 MLD. Raw Narmada water enters with TDS and hardness below 250 mg/L and passes through four steps:

1. **Clariflocculator:** polyaluminium chloride (PAC) is dosed to destabilise suspended particles so turbidity settles out.
2. **Rapid sand filters:** graded sand and gravel remove colloidal matter; the filters are cleaned by periodic backwashing.
3. **Disinfection:** an ALOTC GP chlorination system holds residual chlorine at 0.2 mg/L.
4. **Distribution:** BIS-compliant drinking water (turbidity below 1 NTU) goes to the Water Service Centres (WSC).

Backwash water from the filters is reused rather than discarded: about 9 ML in FY 2025–26.

Rainwater Harvesting: Jal Mandaps

Five underground Jal Mandaps, inspired by Gujarat's traditional vav (stepwells), store rooftop rainwater and double as landscape features where people gather.

- **Catchment:** rooftop catchment was expanded from 56,056 m² to 251,000 m².
- **Monsoon use:** harvested rainwater is mixed with turbid Narmada canal water to dilute it, which eases treatment at the WTP. Harvesting also cuts the energy needed to pump municipal water.
- **Irrigation:** about 1,370 kL of harvested water went to irrigation in FY 2025–26.

Jal Mandap	Storage capacity
Jal Mandap 1	1,400 kL
Jal Mandap 2	1,400 kL
Jal Mandap 3	1,400 kL
Jal Mandap 4	628 kL
Jal Mandap 5	1,400 kL
Total	6,228 kL

Performance. The network collected 24.8 ML in 2024 at 92% system efficiency, and about 30 ML in FY 2025–26.

Natural Lakes and Stormwater

Natural depressions on the site are used as surface water storage lakes. Stormwater runoff flows by gravity through intake wells into two seasonal ponds, which help recharge groundwater and reduce runoff; their overflow is channelled to the Sabarmati River. The ponds now attract several migratory bird species and have noticeably enriched campus biodiversity.

Wastewater Treatment: DEWATS

All campus sewage is treated on site by a Decentralised Wastewater Treatment System (DEWATS) that uses no energy and no chemicals, and no sewage line leaves the campus.

Feature	Detail
Capacity	1.2 MLD (two plants of 0.6 MLD each), currently treating about 0.8 MLD
Energy	Gravity-fed biological and physical processes only
Chemicals	None in the core treatment cycle
Discharge	No outgoing municipal sewage line; all treated water stays on campus

Treatment stages.

1. **Settler (primary):** sewage enters at low velocity; heavy particles settle, cutting suspended solids and biological load.
2. **Anaerobic baffled reactor (secondary):** wastewater flows up and down through baffled chambers, where anaerobic microbes break down organics and produce biogas.
3. **Planted gravel filter (polishing):** a constructed wetland where biofilms on graded gravel remove residual organics and plant roots supply oxygen.
4. **Pressure sand filter / UF (tertiary):** pressurised filtration removes fine particles and turbidity so the water is safe to reuse.

The role of *Canna indica*. The planted filter beds use *Canna indica*, whose roots do four jobs:

- **Oxygenation:** fibrous roots carry oxygen to the root zone, creating micro-aerobic pockets that help microbes break down pathogens.
- **Nitrogen removal:** nitrifying bacteria convert ammonium to nitrite and then nitrate, and denitrification zones release it as harmless nitrogen gas.
- **Nutrient and metal uptake:** roots absorb excess phosphorus and accumulate heavy metals such as lead, cadmium and chromium.
- **Filtration:** the dense root mass traps suspended solids and improves clarity.

Reuse for Irrigation and Dual Plumbing

Treated wastewater is currently used only for landscape irrigation. About 550 kL a day (about 197.5 ML a year) of DEWATS-treated water goes to campus landscaping through drip and sprinkler networks, so no fresh Narmada water is used for greenery. The precision delivery saves a further 45 ML a year.

Dual plumbing. Buildings have a second pipe network so that recycled water can supply toilet flushing in future; it is installed but not yet in use.

Network	Water carried	Used for	Status
Blue (domestic supply)	WTP-treated Narmada and Jal Mandap water	Showers, taps and drinking fountains	In use
Pink (recycled supply)	DEWATS-treated reclaimed water	Toilet flushing	Installed, not yet in use

Conservation Measures and Smart Monitoring

Demand is kept down by efficient fixtures, and losses are caught early by metering and automation.

In buildings.

- **Low-flow fixtures:** water-efficient showers, taps and flushes.
- **RO-free campus:** eliminating reverse osmosis units across campus saves about 15 ML a year.

Monitoring and control.

- **IoT tracking:** a campus-wide network of flow meters and level indicators tracks the fresh, harvested and recycled lines through a central monitoring app.
- **VFD pumps:** variable frequency drive pumps match output to demand and cut energy use.
- **Overflow prevention:** sensor-based motorised solenoid valves shut off supply automatically before overhead tanks overflow.
- **Alarms:** smart meters and level indicators raise alarms for prompt action.
- **Leak response:** a campus-wide digital complaint system routes reported pipeline leaks for immediate repair.

Water Savings, FY 2025–26

The four main measures together saved about 266.5 ML in FY 2025–26.

Measure	Water saved or reused (ML)
DEWATS-treated water used for irrigation	197.5
Drip and sprinkler irrigation	45
RO-free campus	15
WTP backwash water reused	9
Total	266.5